The Connes–Størmer entropy of two maximal abelian subalgebras: a closed form for M_2 and partitions that beat bases in M_n

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Abstract

The Connes–Størmer relative entropy $H(P|Q)$ measures the relative position of two subalgebras P, Q of a finite von Neumann algebra. Its simplest non-trivial instance is a pair of maximal abelian subalgebras of $M_n(\mathbb{C})$, the diagonal algebra Δ_n and a rotated copy $U\Delta_n U^*$. Choda's value $h(U)$, the averaged row entropy of the unistochastic matrix $(|u_{ij}|^2)$, is a lower bound, and Petz, Szántó and Weiner showed that it is not the entropy already for $n = 2$; a formula for $H(\Delta_2|U\Delta_2U^*)$ has been described as open by them and by Bakshi and Guin. We prove that for $U \in U(2)$ with $|u_{11}|^2 = \cos^2\theta$,

$$H(\Delta_2|U\Delta_2U^*) = \max_{\alpha\in\mathbb{R}} \int_{\alpha-\theta}^{\alpha} \sin(2\beta)\log\tan^2\beta\, d\beta,$$

attained by an explicit orthonormal basis whose angle we locate. Consequently $H \le \kappa \min(\theta, \frac{\pi}{2}-\theta)$ with $\kappa = \max_\beta \sin(2\beta)\log\tan^2\beta \approx 1.3255$, and $H/\theta \to \kappa$ as $\theta \to 0$, whereas Choda's value is $o(\theta)$. For arbitrary subalgebras of M_n we note the bound $H \le G^*$ by the maximum of the entropy difference over states, with equality exactly when $\frac{1}{n}$ is a convex combination of maximisers; it is attained for $n = 2$. For $n = 3, 4, 5$ a batched gradient search over partitions of unity and over orthonormal bases, with high-precision recomputation of the best points, shows a different picture: for a small fraction of Haar-random U the best partition found, with $n+1$ rank-one elements, beats every orthonormal basis found, and in all samples the best value found lies strictly between $h(U)$ and the computed G^*. We state these findings as observations and conjectures.

Keywords: Connes–Størmer entropy; relative entropy of subalgebras; maximal abelian subalgebras; unistochastic matrices; partitions of unity; complementary bases; numerical optimisation.
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1 Introduction

Let $\mathcal{M}$ be a finite von Neumann algebra with a faithful normal tracial state τ, let $\eta(t) = -t\log t$, and for a von Neumann subalgebra $P \subset \mathcal{M}$ let E_P denote the τ-preserving conditional expectation onto P. The relative entropy of Connes and Størmer [3] of two subalgebras P, Q is

$$H(P|Q) = \sup \sum_j \big[\tau\eta(E_Q x_j) - \tau\eta(E_P x_j)\big], \tag{1}$$

the supremum being taken over all finite partitions of unity $x_j \ge 0$, $\sum_j x_j = 1$, in $\mathcal{M}$. Connes and Størmer introduced it for finite-dimensional subalgebras on the way to the entropy of automorphisms [3]; Pimsner and Popa extended it to arbitrary subalgebras and related the entropy $H(\mathcal{M}|N)$ of an inclusion $N \subset \mathcal{M}$ to the Jones index [11, 6]. The partition $x_1 = 1$ shows $H(P|Q) \ge 0$, and $H(P|P) = 0$. As a measure of the relative position of two subalgebras (cf. [1]), its simplest non-trivial instance is a pair of maximal abelian subalgebras (masas) of $M_n = M_n(\mathbb{C})$ with $\tau = \mathrm{Tr}\,/n$: the diagonal algebra Δ_n and a rotated copy $U\Delta_nU^*$, U unitary.

The partition into the minimal projections e_{jj} of Δ_n gives the lower bound

$$H(\Delta_n|U\Delta_nU^*) \;\geq\; h(U) = \frac{1}{n}\sum_{i,j}\eta\big(|u_{ij}|^2\big), \tag{2}$$

the entropy of the unistochastic matrix $(|u_{ij}|^2)$, averaged over its rows. Choda [2] showed that $h(U)$ is the value of a modified relative entropy in which the partitions are taken in Δ_n. The expectation that $H(\Delta_n|U\Delta_nU^*) = h(U)$ was disproved for $n = 2$ by Petz, Szántó and Weiner, who remarked that the correct formula "seems to be a hard problem" [10, Appendix]. Bakshi and Guin bound the Connes–Størmer entropy of pairs of spin model subfactors built from 2×2 complex Hadamard matrices from below by $h(U)$ and note that a formula for $H(\Delta_n|U\Delta_nU^*)$ is open even for $n = 2$ [1, Section 2.3 and Theorem 6.22]. At the other extreme, for complementary masas ($|u_{ij}|^2 = \frac{1}{n}$ for all i, j, e.g. the Fourier matrix) one has $h(U) = \log n$, which is the largest possible value, and $H = \log n$ holds only for complementary masas [10, Theorem 3] (see also [4, Theorem 4.11]).

Results. Write $h(p) = \eta(p) + \eta(1-p)$ for the binary entropy and

$$\psi(\beta) = \sin(2\beta)\log\tan^2\beta \qquad (\beta \in \mathbb{R}), \tag{3}$$

extended by continuity with $\psi = 0$ on $\frac{\pi}{2}\mathbb{Z}$. Every unitary $U \in M_2$ satisfies $|u_{11}| = |u_{22}| = \cos\theta$ for a unique $\theta \in [0, \frac{\pi}{2}]$.

Theorem A. *Let $U \in M_2$ be unitary, $c = |u_{11}|^2 = \cos^2\theta$ with $\theta \in [0, \frac{\pi}{2}]$, and*

$$\varphi(\alpha) = h\big(\cos^2(\alpha-\theta)\big) - h(\cos^2\alpha) = \int_{\alpha-\theta}^{\alpha}\psi(\beta)\,d\beta.$$

Then $H(\Delta_2|U\Delta_2U^*) = \max_{\alpha\in\mathbb{R}}\varphi(\alpha)$. The maximum is attained by a partition into the two projections onto an explicit orthonormal basis of $\mathbb{C}^2$, determined by a maximiser α^* of φ and the phases of U. For $\frac{1}{2} < c < 1$ the maximiser is unique modulo $\frac{\pi}{2}$, lies in $(\theta - \frac{\pi}{4}, 0)$ and is the unique solution there of $\psi(\alpha) = \psi(\alpha - \theta)$; H is invariant under $c \mapsto 1 - c$ (Theorem 4.4 and Proposition 4.6).

Theorem B. *(i) For unital $*$-subalgebras $P, Q \subset M_n$ let $G(\rho) = S(E_Q\rho) - S(E_P\rho)$ on the states ρ of M_n, with the von Neumann entropy S, and $G^* = \max_\rho G(\rho)$. Then $H(P|Q) \leq G^*$, with equality if and only if $\frac{1}{n}$ is a convex combination of maximisers of G (Proposition 3.1). For two masas of M_2 equality holds.*
(ii) For $n = 2$, with the notation of Theorem A, $h(c) \leq H(\Delta_2|U\Delta_2U^*) \leq \kappa\min(\theta, \frac{\pi}{2} - \theta)$, where $\kappa = \max\psi = 2\sqrt{1-y^2}/y \approx 1.3255$ and $y \in (0,1)$ is the unique solution of $y\,\mathrm{artanh}\,y = 1$. As $\theta \to 0$,

$$H(\Delta_2|U\Delta_2U^*) = \kappa\theta - \frac{\kappa + 4/\kappa}{6}\,\theta^3 + O(\theta^4), \qquad h(c) = o(\theta),$$

so $H/h(U) \to \infty$ as $c \to 1$ (Theorem 5.1).

For $n \geq 3$ we have no closed form. We computed $H(\Delta_n|U\Delta_nU^*)$ for Haar-random U with $n = 3, 4, 5$ by batched multistart gradient ascent over partitions of unity with n^2 elements and, separately, over orthonormal bases, with the PyTorch library `torch_vn_algebra` [7] (Section 6). The search reproduces Theorem A for $n = 2$ to 5×10^{-13} and shows:

Observation C. For $n = 3, 4, 5$ the supremum (1) for two masas is in general not attained at an orthonormal basis: for 22/300, 6/150 and 2/24 of the sampled unitaries a partition with $n+1$ rank-one elements beats the best basis found, by up to 6.9×10^{-3} ($n = 3$); the partition values are recomputed with 50 digits at explicit points. In all samples the best value found exceeds $h(U)$ and stays below the computed G^*: Choda's partition is not optimal, and the bound of Theorem B(i) does not appear to be attained.

Since no upper bound over all bases is certified, Observation C is not a theorem; we formulate the corresponding conjectures in Section 6 and list open problems in Section 7.

What is new. The ensemble form of H (Proposition 2.3), the bound $H \le \log n$, the strict inequality $H(\Delta_2|U\Delta_2U^*) > h(U)$ for $c \notin \{0, \frac{1}{2}, 1\}$ and the characterisation of $H = \log n$ are due to Petz, Szántó and Weiner [10]; the lower bound (2) and its interpretation are Choda's [2]. Theorem A, the bounds of Theorem B and Observation C are new. Theorem B(i) is an elementary consequence of the ensemble form; we state it because its equality case separates $n = 2$ from $n \ge 3$. A short account of part of these results appears as a use case in the description of the software used for the computations [7].

Organisation. Section 2 collects definitions and the ensemble form. Section 3 proves Theorem B(i), Section 4 Theorem A and Section 5 Theorem B(ii). Section 6 describes the computations for $n \ge 3$ and Section 7 lists open problems.

2 Preliminaries

2.1 Conditional expectations and the entropy

Throughout, $\mathcal{M} = M_n$, $\tau = \operatorname{Tr}/n$, and $\langle x, y\rangle_\tau = \tau(x^*y)$. Subalgebras are unital $*$-subalgebras. For such a subalgebra P the τ-preserving conditional expectation E_P is the orthogonal projection of $(\mathcal{M}, \langle\cdot,\cdot\rangle_\tau)$ onto P; it is unital, positive, P-bimodular, preserves τ and Tr, and is the unique conditional expectation onto P with these properties (see e.g. [12]). The function $\eta(t) = -t\log t$, $\eta(0) = 0$, is continuous and concave on $[0,\infty)$; for $x \ge 0$, $\eta(x)$ is defined by functional calculus, and $S(\sigma) = \operatorname{Tr}\eta(\sigma)$ is the von Neumann entropy of a density matrix σ. The state space $\mathcal{D} = \{\rho \in M_n : \rho \ge 0,\ \operatorname{Tr}\rho = 1\}$ is a compact convex set of real dimension $n^2 - 1$.

Definition 2.1. For subalgebras $P, Q \subset \mathcal{M}$ and $x \ge 0$ put $F(x) = F_{P,Q}(x) = \tau\eta(E_Qx) - \tau\eta(E_Px)$. The Connes–Størmer relative entropy is $H(P|Q) = \sup\sum_j F(x_j)$ over all finite families $x_1, \dots, x_m \ge 0$ in $\mathcal{M}$ with $\sum_j x_j = 1$ (partitions of unity).

Lemma 2.2. *Let $P, Q \subset \mathcal{M}$ be subalgebras.*

(a) $F(\lambda x) = \lambda F(x)$ for $x \ge 0$ and $\lambda \ge 0$. In particular, merging two elements of a partition that are multiples of the same positive operator does not change $\sum_j F(x_j)$.

(b) Let $v \in \mathcal{M}$ be unitary and $\gamma = \operatorname{Ad} v$. Then $E_{\gamma(R)} = \gamma \circ E_R \circ \gamma^{-1}$ for every subalgebra R, $F_{\gamma(P),\gamma(Q)}(\gamma(x)) = F_{P,Q}(x)$, and $H(\gamma(P)|\gamma(Q)) = H(P|Q)$; γ maps optimal partitions for (P, Q) onto optimal partitions for $(\gamma(P), \gamma(Q))$.

Proof. (a) For $s, \lambda \ge 0$, $\eta(\lambda s) = \lambda\eta(s) + s\,\eta(\lambda)$, hence $\tau\eta(\lambda y) = \lambda\tau\eta(y) + \eta(\lambda)\tau(y)$ for $y \ge 0$. Since $\tau(E_Qx) = \tau(E_Px) = \tau(x)$, the terms with $\eta(\lambda)$ cancel in $F(\lambda x)$. If $x_1 = a\,y$ and $x_2 = b\,y$ with $a, b \ge 0$, then $F(x_1) + F(x_2) = (a+b)F(y) = F(x_1 + x_2)$.

(b) γ is unitary for $\langle\cdot,\cdot\rangle_\tau$, since $\tau(\gamma(x)^*\gamma(y)) = \tau(vx^*yv^*) = \tau(x^*y)$, and $\gamma(R)$ is the image of R; so $\gamma E_R\gamma^{-1}$ is the orthogonal projection onto $\gamma(R)$. Moreover $\eta(\gamma(y)) = \gamma(\eta(y))$ and $\tau \circ \gamma = \tau$, so $\tau\eta(E_{\gamma(R)}\gamma(x)) = \tau\gamma(\eta(E_Rx)) = \tau\eta(E_Rx)$. Since γ maps partitions of unity bijectively onto partitions of unity, the remaining claims follow. □

2.2 Two masas

Let Δ_n be the diagonal masa and $Q_U = U\Delta_n U^*$ for a unitary U with columns $u_1, \dots, u_n$. The minimal projections of Q_U are $p_k = u_k u_k^*$, and

$$E_{\Delta_n}(x) = \sum_k e_{kk} x e_{kk}, \qquad E_{Q_U}(x) = \sum_k p_k x p_k = \sum_k \langle u_k, x u_k\rangle\, p_k, \tag{4}$$

since $\sum_k p_k x p_k \in Q_U$ and $\tau\big(p_l(x - \sum_k p_k x p_k)\big) = \tau(p_l x) - \tau(p_l x p_l) = 0$ for every l. Thus the spectrum of $E_{Q_U} x$ is the diagonal of $U^* x U$, and

$$F(x) = \frac{1}{n}\Big[\sum_k \eta\big((U^*xU)_{kk}\big) - \sum_k \eta(x_{kk})\Big]. \tag{5}$$

The algebra Q_U does not change if U is multiplied from the right by a diagonal unitary or a permutation matrix, and by Lemma 2.2(b) with $\gamma = \operatorname{Ad} D$ for a diagonal unitary D (which fixes Δ_n), $H(\Delta_n|Q_{DU}) = H(\Delta_n|Q_U)$. For the partition $(e_{kk})_k$, $\eta(x_{kk}) \in \{\eta(0), \eta(1)\} = \{0\}$ and $(U^* e_{kk} U)_{jj} = |u_{kj}|^2$, which gives (2). For an orthonormal basis $w_1, \dots, w_n$, the columns of a unitary W, the partition $x_j = w_j w_j^*$ gives, by (5),

$$\sum_j F(w_j w_j^*) = h(U^*W) - h(W), \tag{6}$$

and Choda's partition is the case $W = 1$.

2.3 The ensemble form

The following reformulation is due to Petz, Szántó and Weiner [10, Section 3, Lemma 1 and (7), (8)]; we include the short proof, and a Carathéodory argument for the attainment of the supremum.

Proposition 2.3 (Petz–Szántó–Weiner). *Let $P, Q \subset M_n$ be subalgebras and $G(\rho) = S(E_Q\rho) - S(E_P\rho)$ for $\rho \in \mathcal{D}$.*

(a) A partition (x_j) with $x_j \neq 0$ corresponds to the ensemble $t_j = \tau(x_j)$, $\rho_j = x_j/\operatorname{Tr} x_j$, which satisfies $t_j > 0$, $\sum_j t_j = 1$ and $\sum_j t_j \rho_j = \frac{1}{n}$; conversely every such ensemble defines the partition $x_j = n t_j \rho_j$. In both directions $\sum_j F(x_j) = \sum_j t_j G(\rho_j)$.

(b) $H(P|Q) = \max\big\{\sum_j t_j G(\rho_j) :\ t_j \geq 0,\ \sum_j t_j = 1,\ \sum_j t_j \rho_j = \frac{1}{n}\big\}$; the maximum is attained by an ensemble of at most $n^2 + 1$ states. (Petz, Szántó and Weiner show that n^2 linearly independent states suffice.)

(c) $G \leq \log n$ on $\mathcal{D}$, hence $H(P|Q) \leq \log n$.

Proof. (a) Elements $x_j = 0$ contribute $F(0) = 0$ and can be dropped. $\operatorname{Tr} x_j = n\tau(x_j) = n t_j$, so $x_j = n t_j \rho_j$, and $\sum_j x_j = 1$ is equivalent to $\sum_j t_j \rho_j = \frac{1}{n}$, which implies $\sum_j t_j = 1$ by taking traces. Since $E_P\rho$ and $E_Q\rho$ are density matrices, $F(\rho) = \frac{1}{n}[S(E_Q\rho) - S(E_P\rho)] = \frac{1}{n}G(\rho)$, and Lemma 2.2(a) gives $F(x_j) = n t_j F(\rho_j) = t_j G(\rho_j)$.

(b) By (a), H is the supremum of $\sum_j t_j G(\rho_j)$ over finite ensembles with barycentre $\frac{1}{n}$. Identify the affine hull of $\mathcal{D}$ with $\mathbb{R}^{n^2-1}$. For such an ensemble the point $(\frac{1}{n}, \sum_j t_j G(\rho_j)) \in \mathbb{R}^{n^2}$ is a convex combination of points of the graph $\{(\rho, G(\rho)) : \rho \in \mathcal{D}\}$; by Carathéodory's theorem it is a convex combination of at most $n^2 + 1$ of them, i.e. there is an ensemble of at most $n^2 + 1$ states with the same barycentre and the same value. The ensembles of $n^2 + 1$ states (zero weights allowed) with barycentre $\frac{1}{n}$ form a compact set, on which $\sum_j t_j G(\rho_j)$ is continuous, because S is continuous on $\mathcal{D}$ and E_P, E_Q are linear. So the supremum is a maximum.

(c) $S(E_Q\rho) \leq \log n$ for the density matrix $E_Q\rho$ of M_n, and $S(E_P\rho) \geq 0$. □

For two masas, $G(\rho) = H_{\mathrm{Sh}}\big(\mathrm{diag}(U^*\rho U)\big) - H_{\mathrm{Sh}}\big(\mathrm{diag}\,\rho\big)$ with the Shannon entropy H_{Sh} of a probability vector, by (4).

3 The concave envelope and the bound G^*

For a bounded function g on $\mathcal{D}$ let $\hat{g}(\rho) = \sup\{\sum_j t_j g(\rho_j) :$ finite ensembles with $\sum_j t_j\rho_j = \rho\}$; this is the smallest concave function on $\mathcal{D}$ that majorises g (its concave envelope).

Proposition 3.1. *Let $P, Q \subset M_n$ be subalgebras, G as in Proposition 2.3, $G^* = \max_{\mathcal{D}} G$ and $A = \{\rho \in \mathcal{D} : G(\rho) = G^*\}$. Then*

$$H(P|Q) = \hat{G}\big(\tfrac{1}{n}\big) \le G^*,$$

and $H(P|Q) = G^*$ if and only if $\frac{1}{n} \in \operatorname{conv} A$. In that case the optimal partitions are exactly the partitions $x_j = nt_j\rho_j$ with $\rho_j \in A$ whenever $t_j > 0$.

Proof. The identity $H = \hat{G}(\frac{1}{n})$ is Proposition 2.3(b) with the definition of $\hat{G}$. For every ensemble, $\sum_j t_j G(\rho_j) \le \sum_j t_j G^* = G^*$, with equality if and only if $G(\rho_j) = G^*$ whenever $t_j > 0$. If $\frac{1}{n} = \sum_j t_j\rho_j$ with $\rho_j \in A$, this ensemble has the value G^*, so $H = G^*$. Conversely, if $H = G^*$, the maximum in Proposition 2.3(b) is attained by an ensemble with value G^*, all of whose states with positive weight lie in A; hence $\frac{1}{n} \in \operatorname{conv} A$. The last statement is the equality case together with Proposition 2.3(a). □

Remark 3.2. (i) G^* is a maximisation over the (n^2-1)-dimensional state space, whereas H requires ensembles of up to n^2 states with a constraint; numerically G^* is much cheaper. Any state gives a lower bound for G^*, any partition a lower bound for H. (ii) For two masas of M_2, $H = G^*$ (Theorem 4.4). For $n = 3, 4, 5$ the computations of Section 6 find $G^* > H$ for every sampled unitary, i.e. the optimal ensembles do not consist of maximisers of G. (iii) If U is complementary, $G(e_{11}) = \log n = G^*$ by (5), and $\frac{1}{n} = \frac{1}{n}\sum_k e_{kk}$ is a convex combination of the maximisers e_{kk} (all of which satisfy $G(e_{kk}) = \log n$); so $H = G^* = \log n$, the partition (e_{kk}) is optimal, and $h(U) = \log n$.

4 Two masas of M_2

4.1 The function ψ

Let $\Psi(\beta) = h(\cos^2\beta)$ and let ψ be as in (3).

Lemma 4.1. *(a) ψ is continuous on $\mathbb{R}$, odd, of period $\frac{\pi}{2}$, vanishes exactly on $\frac{\pi}{4}\mathbb{Z}$, and $\psi > 0$ on $(-\frac{\pi}{4}, 0)$, $\psi < 0$ on $(0, \frac{\pi}{4})$.*

(b) Ψ is continuously differentiable on $\mathbb{R}$, $\Psi' = -\psi$, and Ψ is even and of period $\frac{\pi}{2}$.

(c) On $(-\frac{\pi}{4}, 0)$, ψ has a unique critical point β^*; ψ is strictly increasing on $[-\frac{\pi}{4}, \beta^*]$ and strictly decreasing on $[\beta^*, 0]$. With $\beta^* = -s^*/2$ and $y = \cos s^*$, $y \in (0,1)$ is the unique solution of $y\,\mathrm{artanh}\, y = 1$, and

$$\kappa := \max_{\mathbb{R}} \psi = \psi(\beta^*) = 2\tan s^* = \frac{2\sqrt{1-y^2}}{y}, \qquad \psi''(\beta^*) = -4\kappa - \frac{16}{\kappa}.$$

Proof. (a) For $\beta \notin \frac{\pi}{2}\mathbb{Z}$, $\psi(-\beta) = -\psi(\beta)$ since sin is odd and $\tan^2$ even, and $\psi(\beta + \frac{\pi}{2}) = \psi(\beta)$ since $\sin(2\beta + \pi) = -\sin 2\beta$ and $\tan^2(\beta + \frac{\pi}{2}) = \cot^2\beta$, so that $\log\tan^2(\beta + \frac{\pi}{2}) = -\log\tan^2\beta$. Near 0, $|\psi(\beta)| \le 2|\beta|\,|\log\tan^2\beta| \to 0$, so ψ is continuous at 0 with $\psi(0) = 0$, and by periodicity at every point of $\frac{\pi}{2}\mathbb{Z}$. On $(0, \frac{\pi}{4})$, $\sin 2\beta > 0$ and $0 < \tan^2\beta < 1$, so $\psi < 0$; by oddness $\psi > 0$ on $(-\frac{\pi}{4}, 0)$. $\psi(\pm\frac{\pi}{4}) = \pm\log 1 = 0$. By periodicity these are all zeros.

(b) h is smooth on $(0,1)$ with $h'(p) = \log\frac{1-p}{p}$. For $\beta \notin \frac{\pi}{2}\mathbb{Z}$, $0 < \cos^2\beta < 1$ and $\Psi'(\beta) = h'(\cos^2\beta)\cdot(-\sin 2\beta) = -\sin 2\beta \log\tan^2\beta = -\psi(\beta)$. At $\beta_0 \in \frac{\pi}{2}\mathbb{Z}$, Ψ is continuous, and by the mean value theorem $(\Psi(\beta)-\Psi(\beta_0))/(\beta-\beta_0) = -\psi(\xi)$ for some ξ between β_0 and β; by continuity of ψ this tends to $-\psi(\beta_0) = 0$. Hence $\Psi \in C^1(\mathbb{R})$, $\Psi' = -\psi$. Evenness and periodicity follow from $\cos^2(-\beta) = \cos^2\beta$ and $h(\cos^2(\beta+\frac{\pi}{2})) = h(\sin^2\beta) = h(\cos^2\beta)$.

(c) Away from $\frac{\pi}{2}\mathbb{Z}$, $(\log\tan^2\beta)' = 2/(\sin\beta\cos\beta) = 4/\sin 2\beta$, hence

$$\psi'(\beta) = 2\cos 2\beta \log\tan^2\beta + 4, \qquad \psi''(\beta) = -4\psi(\beta) + 8\cot 2\beta.$$

For $b \in (0, \frac{\pi}{4})$ let $g(b) = \cos 2b \log\tan b$, so that $\psi'(-b) = 4(g(b)+1)$. Then $g'(b) = -2\sin 2b\log\tan b + 2\cos 2b/\sin 2b > 0$, $g(b) \to -\infty$ as $b \to 0$ and $g(\frac{\pi}{4}) = 0$. Hence $g = -1$ at exactly one point b^*, $\psi' > 0$ on $(-\frac{\pi}{4}, -b^*)$ and $\psi' < 0$ on $(-b^*, 0)$; put $\beta^* = -b^*$, $s^* = 2b^* \in (0, \frac{\pi}{2})$. Since $\log\tan b = -\operatorname{artanh}(\cos 2b)$ (both sides equal $\frac{1}{2}\log\frac{1-\cos 2b}{1+\cos 2b}$), the equation $g(b^*) = -1$ reads $y\operatorname{artanh} y = 1$ with $y = \cos s^*$, and $y\operatorname{artanh} y$ increases strictly from 0 to ∞ on $(0,1)$. Then $\psi(\beta^*) = -\sin s^* \cdot 2\log\tan b^* = 2\sin s^*/\cos s^* = 2\tan s^*$. By (a) the maximum of ψ over a period is attained in $[-\frac{\pi}{4}, 0]$, hence at β^*. Finally $\psi''(\beta^*) = -4\kappa + 8\cot(-s^*) = -4\kappa - 8/\tan s^* = -4\kappa - 16/\kappa$. □

Numerically $\kappa = 1.3255$ (computed by `scripts/connes_stormer.py`; the test suite checks $\kappa = 1.3254868$ to 10^{-6}).

4.2 The closed form

Write $R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$.

Lemma 4.2. *Every unitary $U \in M_2$ can be written as $U = D_1 R_\theta D_2$ with diagonal unitaries D_1, D_2 and $\theta \in [0, \frac{\pi}{2}]$, $\cos^2\theta = |u_{11}|^2$. Consequently $H(\Delta_2 | U\Delta_2 U^*) = H(\Delta_2 | R_\theta \Delta_2 R_\theta^T)$, and $\gamma = \operatorname{Ad} D_1$ maps optimal partitions for R_θ onto optimal partitions for U.*

Proof. The columns and the rows of U are unit vectors, so $|u_{11}|^2 + |u_{21}|^2 = 1 = |u_{11}|^2 + |u_{12}|^2$ and $|u_{12}|^2 + |u_{22}|^2 = 1$; hence $|u_{21}| = |u_{12}|$ and $|u_{22}| = |u_{11}|$. Let $\theta \in [0, \frac{\pi}{2}]$ with $\cos\theta = |u_{11}|$ and write $u_{11} = e^{ia}\cos\theta$, $u_{21} = e^{ib}\sin\theta$, $u_{12} = -e^{ic}\sin\theta$, $u_{22} = e^{id}\cos\theta$. If $0 < \theta < \frac{\pi}{2}$, orthogonality of the columns gives $(e^{i(d-b)} - e^{i(c-a)})\cos\theta\sin\theta = 0$, i.e. $e^{i(d-b)} = e^{i(c-a)}$, and then $U = \operatorname{diag}(e^{ia}, e^{ib})\, R_\theta \operatorname{diag}(1, e^{i(c-a)})$, as one checks entry by entry. If $\theta = 0$, U is diagonal and $U = U \cdot R_0 \cdot 1$; if $\theta = \frac{\pi}{2}$, $U = \operatorname{diag}(-u_{12}, u_{21}) R_{\pi/2}$. Since $D_2\Delta_2 D_2^* = \Delta_2$, $U\Delta_2 U^* = \gamma(R_\theta \Delta_2 R_\theta^T)$ with $\gamma = \operatorname{Ad} D_1$, which fixes Δ_2; Lemma 2.2(b) gives the rest. □

We use the Pauli matrices $\sigma = (\sigma_1, \sigma_2, \sigma_3)$ and the Bloch parametrisation $\rho = \frac{1}{2}(1 + r\cdot\sigma)$, $r \in \mathbb{R}^3$, $|r| \le 1$, of $\mathcal{D}$ for $n = 2$; ρ is a rank-one projection if and only if $|r| = 1$. For a unit vector $v = (\cos a, \sin a)^T$ one has $vv^* = \frac{1}{2}(1 + \sin 2a\,\sigma_1 + \cos 2a\,\sigma_3)$, so its Bloch vector is $(\sin 2a, 0, \cos 2a)$, and the Bloch vector of $(-\sin a, \cos a)^T$ is the negative of it. Let

$$z = (0,0,1), \qquad q = (\sin 2\theta, 0, \cos 2\theta), \qquad k(s) = h\big(\tfrac{1+s}{2}\big)\ (|s| \le 1),$$

so z and q are the Bloch vectors of e_1 and of $R_\theta e_1$, and k is even.

Lemma 4.3. *For $P = \Delta_2$, $Q = R_\theta \Delta_2 R_\theta^T$ and $\rho = \frac{1}{2}(1 + r\cdot\sigma)$, $G(\rho) = f(r) := k(r\cdot q) - k(r\cdot z)$. Moreover*

$$k(s) = \log 2 - \sum_{m\ge 1} \frac{s^{2m}}{2m(2m-1)} \qquad (|s| \le 1), \tag{7}$$

the series converging uniformly on $[-1, 1]$.

Proof. $E_{\Delta_2}\rho = \operatorname{diag}(\rho)$ has the eigenvalues $\frac{1}{2}(1 \pm r_3) = \frac{1}{2}(1 \pm r \cdot z)$, so $S(E_{\Delta_2}\rho) = k(r \cdot z)$. By (4) the eigenvalues of $E_Q\rho$ are $\langle v_1, \rho v_1\rangle$ and $\langle v_2, \rho v_2\rangle = 1 - \langle v_1, \rho v_1\rangle$ with $v_1 = R_\theta e_1$, and $\langle v_1, \rho v_1\rangle = \operatorname{Tr}(\rho v_1 v_1^*) = \frac{1}{2}(1 + r \cdot q)$ by $\operatorname{Tr}\big((1 + r \cdot \sigma)(1 + q \cdot \sigma)\big) = 2(1 + r \cdot q)$. So $S(E_Q\rho) = k(r \cdot q)$.

For (7), $k(s) = \log 2 - \frac{1}{2}\big[(1+s)\log(1+s) + (1-s)\log(1-s)\big]$. For $|s| < 1$, integrating $\frac{d^2}{ds^2}(1 \pm s)\log(1 \pm s) = 1/(1 \pm s) = \sum_{j\geq 0}(\mp s)^j$ twice gives $(1 \pm s)\log(1 \pm s) = \pm s + \sum_{l\geq 2}(\mp 1)^l s^l/(l(l-1))$, and adding the two expansions leaves $2\sum_{m\geq 1} s^{2m}/(2m(2m-1))$. The terms are bounded by $1/(2m(2m-1))$ on $[-1, 1]$, so the series converges uniformly there (Weierstrass), its sum is continuous, and the identity extends to $s = \pm 1$ by continuity of k. □

Theorem 4.4. *Let $U \in M_2$ be unitary, $U = D_1 R_\theta D_2$ as in Lemma 4.2, $c = |u_{11}|^2 = \cos^2\theta$, and*

$$\varphi(\alpha) = h\big(\cos^2(\alpha - \theta)\big) - h(\cos^2\alpha) = \int_{\alpha-\theta}^{\alpha} \psi(\beta)\, d\beta.$$

Then φ is C^1 and $\frac{\pi}{2}$-periodic, $\varphi'(\alpha) = \psi(\alpha) - \psi(\alpha - \theta)$, and

(a) $H(\Delta_2|U\Delta_2U^*) = \max_{\alpha\in\mathbb{R}} \varphi(\alpha) = G^*$;

(b) for every maximiser α^* of φ, the projections onto the orthonormal basis $D_1(\cos\alpha^*, \sin\alpha^*)^T$, $D_1(-\sin\alpha^*, \cos\alpha^*)^T$ form an optimal partition;

(c) $H(\Delta_2|U\Delta_2U^*)$ depends only on c, and it is invariant under $c \mapsto 1 - c$.

Proof. By Lemma 4.2 we may assume $U = R_\theta$, $D_1 = 1$, and transport optimal partitions by $\operatorname{Ad} D_1$ at the end. The integral form is $\varphi(\alpha) = \Psi(\alpha - \theta) - \Psi(\alpha) = \int_{\alpha-\theta}^{\alpha}\psi$ by Lemma 4.1(b), which also gives the regularity, periodicity and derivative of φ.

Step 1: $H = G^* = \max_{|r|\leq 1} f(r)$. By Lemma 4.3, $G^* = \max_{|r|\leq 1} f$, the maximum of a continuous function on the closed unit ball. Since k is even, $f(-r) = f(r)$; so if r^* maximises f, the states $\rho_\pm = \frac{1}{2}(1 \pm r^* \cdot \sigma)$ both maximise G, and $\frac{1}{2}\rho_+ + \frac{1}{2}\rho_- = \frac{1}{2}\cdot 1$. By Proposition 3.1, $H = G^*$, and the partition $x_\pm = 2 \cdot \frac{1}{2}\rho_\pm = \rho_\pm$ is optimal.

Step 2: reduction to the unit circle in the plane of z and q. If $\theta \in \{0, \frac{\pi}{2}\}$, then $q = \pm z$ and $f \equiv 0$ since k is even; also $\varphi \equiv 0$, because $h(\cos^2(\alpha - \frac{\pi}{2})) = h(\sin^2\alpha) = h(\cos^2\alpha)$. So $H = 0 = \max\varphi$ and every basis is optimal. Let $0 < \theta < \frac{\pi}{2}$. Then z and q span the plane $\{r_2 = 0\}$, and $f(r)$ depends only on the orthogonal projection $a = (r_1, 0, r_3)$ of r onto it, $|a| \leq |r| \leq 1$. Write $a = \varrho(\sin\beta, 0, \cos\beta)$ with $0 \leq \varrho \leq 1$; then $a \cdot z = \varrho\cos\beta$ and $a \cdot q = \varrho\cos(\beta - 2\theta)$, and by (7)

$$f(a) = \sum_{m\geq 1} \frac{\varrho^{2m}}{2m(2m-1)}\big(\cos^{2m}\beta - \cos^{2m}(\beta - 2\theta)\big).$$

If $|\cos\beta| \geq |\cos(\beta - 2\theta)|$, all terms are non-negative and non-decreasing in ϱ, so $f(a) \leq f(\sin\beta, 0, \cos\beta)$. Otherwise all terms are non-positive and $f(a) \leq 0 = f(\sin\theta, 0, \cos\theta)$, the value at $\beta = \theta$, where $\cos\beta = \cos(\beta - 2\theta)$. Hence the maximum of f over the ball equals its maximum over the unit circle of the plane,

$$\max_{|r|\leq 1} f = \max_{\beta\in\mathbb{R}}\big[k(\cos(\beta - 2\theta)) - k(\cos\beta)\big],$$

and it is attained at a unit vector $r^* = (\sin\beta^\circ, 0, \cos\beta^\circ)$.

Step 3: the angle. Since $k(\cos 2\gamma) = h(\frac{1+\cos 2\gamma}{2}) = h(\cos^2\gamma)$, the substitution $\beta = 2\alpha$ turns the right-hand side into $\max_\alpha \varphi(\alpha)$. This proves (a). For a maximiser α^* of φ, $r^* = (\sin 2\alpha^*, 0, \cos 2\alpha^*)$ maximises f, and ρ_+, ρ_- are the projections onto $(\cos\alpha^*, \sin\alpha^*)^T$ and $(-\sin\alpha^*, \cos\alpha^*)^T$; with Step 1 and Lemma 4.2 this proves (b).

(c) φ depends only on θ, i.e. on c. Let $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Then $UJ\Delta_2(UJ)^* = U\Delta_2U^*$ and $|(UJ)_{11}|^2 = |u_{12}|^2 = 1 - c$. □

Remark 4.5. (i) At $\alpha = 0$, $\varphi(0) = h(c)$, the value of Choda's partition e_{11}, e_{22}; indeed $h(U) = h(c)$ for $n = 2$. At $\alpha = \theta$ (the basis of Q) $\varphi(\theta) = -h(c)$. (ii) In the Bloch picture, z and q enclose the angle 2θ, and the optimal Bloch vector r^* has the angle $2\alpha^*$ from z. By Proposition 4.6 below, for $\frac{1}{2} < c < 1$ it lies on the other side of z than q: the optimal partition accepts some entropy of $E_{\Delta_2} x_j$ in exchange for more entropy of $E_Q x_j$. (iii) Let $0 < c < 1$. Since $H = G^*$, the optimal partitions are exactly the families $x_j = 2t_j\rho_j$ with ρ_j maximising G and $\sum_j t_j\rho_j = \frac{1}{2}$ (Proposition 3.1). Every maximiser r of f has $|r| = 1$ and $r_2 = 0$: as $f(r) = H \geq h(c) > 0$, in Step 2 of the proof we are in the case $|\cos\beta| > |\cos(\beta - 2\theta)|$, where every term of the series is strictly increasing in ϱ, and $r_2 \neq 0$ or $|r| < 1$ would give $\varrho < 1$. Since the maximiser of φ is unique modulo $\frac{\pi}{2}$ (Proposition 4.6), the maximisers of G are the two pure states $\rho_\pm$, and the optimal partition is unique up to splitting $\rho_\pm$ into several multiples (Lemma 2.2(a)).

4.3 The optimal angle

Proposition 4.6. *Let $\frac{1}{2} < c < 1$, i.e. $0 < \theta < \frac{\pi}{4}$, and let β^* be as in Lemma 4.1(c). Then φ has a unique maximiser α^* in $(-\frac{\pi}{4}, \frac{\pi}{4}]$; it satisfies*

$$\max\bigl(\beta^*, \theta - \tfrac{\pi}{4}\bigr) < \alpha^* < \min\bigl(\beta^* + \theta, 0\bigr),$$

and it is the unique solution of $\psi(\alpha) = \psi(\alpha - \theta)$ in this interval. In particular $\alpha^* < 0$, and $\alpha^* \to \beta^*$ as $\theta \to 0$. For $0 < c < \frac{1}{2}$ the maximiser is $\alpha^*(c) = -\alpha^*(1 - c)$. For $c = \frac{1}{2}$ the maximiser is $\alpha^* = 0$ (modulo $\frac{\pi}{2}$) and $H = \log 2$.

Proof. Let $I_k = (-\frac{\pi}{4} + k\frac{\pi}{2}, k\frac{\pi}{2})$ and $N_k = (k\frac{\pi}{2}, \frac{\pi}{4} + k\frac{\pi}{2})$, $k \in \mathbb{Z}$; by Lemma 4.1(a), $\psi > 0$ on the I_k and $\psi < 0$ on the N_k, all of length $\frac{\pi}{4}$. Let $0 < \theta \leq \frac{\pi}{4}$ and let $W = [\alpha - \theta, \alpha]$ be a window. If W met two of the I_k, it would contain the closure of some N_k together with points on both sides of it, so its length would exceed $\frac{\pi}{4}$; hence W meets at most one I_k, in an interval J of length at most θ. Choose a window $W' \subset \overline{I_k}$ of length θ containing J. Since $\psi \geq 0$ on $\overline{I_k}$ and $\psi \leq 0$ on $W \setminus J$,

$$\int_W \psi \leq \int_J \psi \leq \int_{W'} \psi,$$

and the first inequality is strict unless $W \setminus J$ is a null set, i.e. unless $W \subset \overline{I_k}$. Hence every maximising window lies in some $\overline{I_k}$; modulo $\frac{\pi}{2}$, $W \subset [-\frac{\pi}{4}, 0]$, i.e. $\alpha \in [\theta - \frac{\pi}{4}, 0]$.

$c = \frac{1}{2}$. For $\theta = \frac{\pi}{4}$ the only such window is $[-\frac{\pi}{4}, 0]$, so $\alpha^* = 0$ and $H = \Psi(-\frac{\pi}{4}) - \Psi(0) = h(\frac{1}{2}) - h(1) = \log 2$.

$\frac{1}{2} < c < 1$. Now $0 < \theta < \frac{\pi}{4}$. At the end points, $\varphi'(0) = \psi(0) - \psi(-\theta) = -\psi(-\theta) < 0$ and $\varphi'(\theta - \frac{\pi}{4}) = \psi(\theta - \frac{\pi}{4}) - \psi(-\frac{\pi}{4}) = \psi(\theta - \frac{\pi}{4}) > 0$, so neither is a maximiser of the C^1 function φ on $\mathbb{R}$, and every maximiser α^* (modulo $\frac{\pi}{2}$) lies in $(\theta - \frac{\pi}{4}, 0)$ and satisfies $\varphi'(\alpha^*) = 0$, i.e. $\psi(\alpha^*) = \psi(\alpha^* - \theta)$ with $\alpha^* - \theta < \alpha^*$ both in $(-\frac{\pi}{4}, 0)$. As ψ is strictly monotone on $[-\frac{\pi}{4}, \beta^*]$ and on $[\beta^*, 0]$ (Lemma 4.1(c)), two distinct points with equal values lie on different sides of β^*: $\alpha^* - \theta < \beta^* < \alpha^*$. On the interval $L = (\max(\beta^*, \theta - \frac{\pi}{4}), \min(\beta^* + \theta, 0))$ we have $\alpha \in (\beta^*, 0)$ and $\alpha - \theta \in (-\frac{\pi}{4}, \beta^*)$, so $\psi(\alpha)$ is strictly decreasing and $\psi(\alpha - \theta)$ strictly increasing in α; hence φ' is strictly decreasing on L and has at most one zero there. A maximiser exists by continuity and periodicity, so it is unique modulo $\frac{\pi}{2}$. As $\theta \to 0$, $L \subset (\beta^*, \beta^* + \theta)$ shrinks to β^*.

$0 < c < \frac{1}{2}$. With $\theta' = \frac{\pi}{2} - \theta$ and Lemma 4.1(b), $\varphi_\theta(\alpha) = \Psi(\alpha - \frac{\pi}{2} + \theta') - \Psi(\alpha) = \Psi(-\alpha - \theta') - \Psi(-\alpha) = \varphi_{\theta'}(-\alpha)$, where φ_θ denotes φ for the angle θ; and $\cos^2\theta' = 1 - c$. □

At $\theta \to 0$ the optimal basis therefore does not approach the basis of Δ_2 (angle 0) or of Q (angle θ), but the fixed angle β^*; the middle panel of Figure 1 shows α^* against c.

5 Bounds for $n = 2$

Theorem 5.1. *Let $U \in M_2$ be unitary, $c = |u_{11}|^2 = \cos^2\theta$, $\theta \in [0, \frac{\pi}{2}]$, and $\kappa = \max\psi$ as in Lemma 4.1(c). Then*

(a) $h(c) \le H(\Delta_2|U\Delta_2U^*) \le \kappa \min\bigl(\theta, \frac{\pi}{2} - \theta\bigr)$;

(b) as $\theta \to 0$, $H(\Delta_2|U\Delta_2U^*) = \kappa\theta - \frac{\kappa+4/\kappa}{6}\theta^3 + O(\theta^4)$, while $h(c) = o(\theta)$; in particular $H/h(U) \to \infty$ as $c \to 1$, and likewise as $c \to 0$.

Proof. (a) The lower bound is $\varphi(0) = h(c)$. For every α, $\varphi(\alpha) = \int_{\alpha-\theta}^{\alpha} \psi \le \theta \max\psi = \kappa\theta$, so $H \le \kappa\theta$ by Theorem 4.4(a). Applying this to $1 - c$, whose angle is $\frac{\pi}{2} - \theta$, and using Theorem 4.4(c) gives $H \le \kappa(\frac{\pi}{2} - \theta)$.

(b) Fix $\delta > 0$ with $[\beta^* - \delta, \beta^* + \delta] \subset (-\frac{\pi}{4}, 0)$; ψ is smooth there, and by Taylor's theorem there is C with $|\psi(\beta) - \kappa - \frac{1}{2}\psi''(\beta^*)(\beta - \beta^*)^2| \le C|\beta - \beta^*|^3$ for $|\beta - \beta^*| \le \delta$. Let $\theta < \min(\delta, \frac{\pi}{4})$. By Proposition 4.6 the maximising window $W = [\alpha^* - \theta, \alpha^*]$ contains β^*, so $W \subset [\beta^* - \delta, \beta^* + \delta]$. Since $\int_W (\beta - \beta^*)^2 d\beta \ge \theta^3/12$ for every window of length θ (with equality for the window centred at β^*), $\psi''(\beta^*) < 0$ and $\int_W |\beta - \beta^*|^3 d\beta \le \theta^4$,

$$\kappa\theta + \frac{\psi''(\beta^*)}{24}\theta^3 - C\theta^4 \;\le\; \varphi\bigl(\beta^* + \tfrac{\theta}{2}\bigr) \;\le\; H \;=\; \int_W \psi \;\le\; \kappa\theta + \frac{\psi''(\beta^*)}{24}\theta^3 + C\theta^4,$$

and $\psi''(\beta^*)/24 = -(\kappa + 4/\kappa)/6$ by Lemma 4.1(c). Finally $h(c) = \varphi(0) = \int_{-\theta}^{0} \psi$ and $|\int_{-\theta}^{0} \psi| \le \theta \max_{[-\theta,0]} |\psi| = o(\theta)$, as ψ is continuous with $\psi(0) = 0$. Since $h(U) = h(c) > 0$ for $0 < c < 1$, $H/h(U) \ge (\kappa\theta + O(\theta^3))/o(\theta) \to \infty$. The case $c \to 0$ follows from Theorem 4.4(c). □

Since $\theta = \arccos\sqrt{c} \sim \sqrt{1-c}$, near $c = 1$ the entropy behaves like $\kappa\sqrt{1-c}$, much larger than $h(c) \sim (1-c)\log\frac{1}{1-c}$. The bound $\log 2$ of Proposition 2.3(c) is better than $\kappa\min(\theta, \frac{\pi}{2} - \theta)$ near $c = \frac{1}{2}$; Table 1 lists $\kappa\theta$ for comparison.

Corollary 5.2 (Petz–Szántó–Weiner)**.** *$H(\Delta_2|U\Delta_2U^*) > h(U)$ unless $c \in \{0, \frac{1}{2}, 1\}$, where $H = h(U)$.*

Proof. This is [10, Appendix]; in terms of Theorem 4.4 it reads as follows. $h(U) = h(c) = \varphi(0)$ and $\varphi'(0) = \psi(0) - \psi(-\theta) = \psi(\theta)$, which by Lemma 4.1(a) vanishes exactly for $\theta \in \{0, \frac{\pi}{4}, \frac{\pi}{2}\}$. Otherwise 0 is not a maximiser of φ, so $H > \varphi(0)$. For $c \in \{0, 1\}$, $H = 0 = h(c)$, and for $c = \frac{1}{2}$, $H = \log 2 = h(\frac{1}{2})$ (Proposition 4.6). □

Remark 5.3 (Spin model subfactors). Bakshi and Guin use the inequality $H(u\Delta_2u^*|v\Delta_2v^*) \le H(R_u|R_v)$ for the spin model subfactors R_u, R_v of two 2×2 complex Hadamard matrices u, v [1, proof of Theorem 6.22], together with Choda's bound. By Lemma 2.2(b) with $\gamma = \mathrm{Ad}\,u^*$, $H(u\Delta_2u^*|v\Delta_2v^*) = H(\Delta_2|U\Delta_2U^*)$ with $U = u^*v$, so Theorem 4.4 with $c = |(u^*v)_{11}|^2$ raises their lower bound $h(c)$ for $H(R_u|R_v)$ to $\max_\alpha \varphi(\alpha)$, e.g. from 0.3251 to 0.4024 at $c = 0.9$ (Table 1).

6 Numerical study for $n \ge 3$

6.1 Method

All computations use our PyTorch [9] library `torch_vn_algebra` [7]; the program is `scripts/connes_stormer.py` of the repository [7], in double precision (complex128) on a CPU.

The functional. Δ_n is the image of the inclusion $\mathbb{C}^n \subset M_n$ with inclusion matrix $(1, \dots, 1)$, and Q_U the image of the same inclusion twisted by U (`inclusions.Inclusion` with `unitary=U`); the

c	$h(c)$	H, Theorem 4.4	H, search	α^*	$\kappa\theta$
0.99	0.056002	0.132043	0.132043	-0.2438	0.132771
0.95	0.198515	0.290618	0.290618	-0.1864	0.298915
0.9	0.325083	0.402396	0.402396	-0.1452	0.426476
0.8	0.500402	0.542607	0.542607	-0.0899	0.614559
0.7	0.610864	0.627949	0.627949	-0.0504	0.768305
0.6	0.673012	0.676658	0.676658	-0.0205	0.907586
0.5	0.693147	0.693147	0.693147	+0.0000	1.041035

Table 1: $H(\Delta_2|U\Delta_2U^*)$ for $c = |u_{11}|^2$: Choda's value $h(c)$, the closed form of Theorem 4.4 with its optimal angle $\alpha^* \in (-\frac{\pi}{4}, \frac{\pi}{4}]$ (radians), the maximum over general partitions with four elements found by the batched search of Section 6 for $U = D_1R_\theta D_2$ with random phases (16 starts), and the upper bound $\kappa\theta$ of Theorem 5.1 (full run on CPU (4 threads)).

conditional expectations onto both are taken for the normalised trace $\tau = \mathrm{Tr}\,/n$, and $\tau\eta(E(x))$ is computed from the eigenvalues of $E(x)$. The whole computation is batched and differentiable. The test suite checks this path against the plain NumPy formula (5) to 10^{-12} for random partitions with $n = 2, 3, 5$.

Search spaces. (1) General partitions with $m = n^2$ elements,

$$x_j = S^{-1/2} Z_j Z_j^* S^{-1/2}, \qquad S = \sum_{j=1}^{m} Z_j Z_j^*,$$

with free complex $Z_j \in M_n$; every family with invertible S is a partition of unity, and every partition of unity with m elements arises this way ($Z_j = x_j^{1/2}$, $S = 1$). By the result of Petz, Szántó and Weiner quoted in Proposition 2.3(b), $m = n^2$ elements suffice. (2) Orthonormal bases, $x_j = w_j w_j^*$ for the columns w_j of a unitary $W = W_0 \exp(X - X^*)$, with value (6). (3) States $\rho = ZZ^*/\,\mathrm{Tr}(ZZ^*)$ for G^*; the reported G^* is the larger of this maximum and the values $G(\rho_j)$ at the normalised elements of the best partition found.

Optimiser. `optimize.extremize` runs Adam with the learning rate $0.05 \cdot 0.6^r$ in rounds $r = 0, \ldots, 7$, re-centring the unitary chart ($W_0 \leftarrow W$, $X \leftarrow 0$) after every round, and returns per start the best value attained at any iterate; every iterate is an admissible point. All starts for all unitaries form one batch. We used 1200 steps for $n = 3, 4$ and 2000 for $n = 5$, and the numbers of starts per unitary given in Table 2 (16 for the states). The unitaries U are Haar-random, from the library's sampler (QR decomposition with phase correction [8]), with fixed seeds: 300 for $n = 3$, 150 for $n = 4$ and 24 for $n = 5$.

Extended basis search. A unitary is flagged if the best partition beats the best basis by more than 1×10^{-8}. For every flagged unitary the basis search is repeated with 64 further starts, half of them Haar-random and half at bases built from the directions of the best partition (the principal eigenvectors of its elements, n of them drawn with probabilities proportional to their traces, perturbed by complex Gaussian noise of size 10^{-3} and orthonormalised by the polar decomposition). For the unitary with the largest remaining gap (for each n) another 1024 starts are run, again half of each kind.

Recomputation in high precision. For every unitary that remains flagged, the best partition and the best basis are recomputed in mpmath with 50 digits: U is replaced by its polar unitarisation $U' = U(U^*U)^{-1/2}$, exactly unitary at this precision, and the partition $x_j = S^{-1/2} Z_j Z_j^* S^{-1/2}$ is re-normalised from the stored Z_j (for the basis, $Z_j = w_j$), so that $|\sum_j x_j - 1|$ is of the order 10^{-50}. The value at such an explicit point is a lower bound for $H(\Delta_n|U'\Delta_n U'^*)$, up to rounding errors far below

n	U	starts	partition $>$ basis	largest gap	agreement otherwise	$H - h(U)$: min, median, max			$G^* > H$
3	300	8/8	22	7×10^{-3}	8×10^{-12}	0.0002	0.0322	0.1333	300
4	150	8/8	6	3×10^{-4}	1×10^{-11}	0.0047	0.0327	0.1762	150
5	24	16/16	2	5×10^{-4}	7×10^{-13}	0.0122	0.0323	0.0849	24

Table 2: Haar-random $U \in U(n)$: starts per unitary for general partitions ($m = n^2$ elements) and for bases; the number of unitaries for which a partition beats the best basis found (after the extended basis search) and the largest such gap (both values recomputed in mpmath); the largest difference $|H_{\text{partitions}} - H_{\text{bases}}|$ for the others; the excess of H (the best value found) over Choda's $h(U)$; and the number of unitaries with $G^* > H + 10^{-6}$ (full run on CPU (4 threads), 726 s in total).

the digits we report. This is a high-precision evaluation, not interval arithmetic; and the basis value is only the best one found, not an upper bound over all bases.

Structure of a partition. Elements of trace below 10^{-7} times the largest trace are discarded; an element counts as rank one if its second eigenvalue is at most 10^{-6} times its trace; the principal eigenvectors d, d' of two elements are merged into one direction if $|\langle d, d' \rangle| > 1 - 10^{-4}$. By Lemma 2.2(a), merging exactly parallel rank-one elements does not change the value.

6.2 Checks

For $n = 2$ the batched search over general partitions with four elements, for $U = D_1 R_\theta D_2$ with random phases, reproduces Theorem 4.4 to 5×10^{-13}, and the optimal basis of Theorem 4.4(b), evaluated in NumPy [5], gives the closed form to 4×10^{-16} (Table 1, Figure 1). The test suite compares the closed form also with a plain BFGS optimisation (SciPy [13]) over partitions of M_2 with three elements for four values of c, checks that 300 random partitions with two to five elements never exceed it, that the value does not depend on the phases of U, the integral representation of φ by quadrature, the Bloch reduction of Lemma 4.3, the ensemble identity of Proposition 2.3(a) for $n = 2, 3, 4$, and the agreement of the mpmath recomputation with the double-precision value. For complementary masas (Fourier matrices, $n = 2, 3, 4$) the search gives $H = \log n$ to 1×10^{-10} (Remark 3.2(iii)), and $H(\Delta_3 | \Delta_3)$ vanishes to 1×10^{-17}.

6.3 Results

Table 2 and Figure 1 summarise the computation for $n = 3, 4, 5$.

(i) *Partitions and bases mostly agree.* For most unitaries the best general partition and the best basis agree to 8×10^{-12}, 1×10^{-11} and 7×10^{-13} for $n = 3, 4, 5$; for $n = 2$ this is Theorem 4.4.

(ii) *But not always.* For 22 of the 300 unitaries with $n = 3$, 6 of the 150 with $n = 4$ and 2 of the 24 with $n = 5$, a partition beats the best basis found by more than 1×10^{-8}. None of them was cleared by the extended basis search, nor by the further 1024 starts for the unitary with the largest gap. The recomputed gaps range from 3×10^{-7} to 6.9×10^{-3} ($n = 3$), from 1×10^{-5} to 2.5×10^{-4} ($n = 4$) and from 1×10^{-6} to 5.3×10^{-4} ($n = 5$). For the unitary with the largest gap the partition gives

$$H(\Delta_3 | U' \Delta_3 U'^*) \geq 0.5727784466, \quad H(\Delta_4 | U' \Delta_4 U'^*) \geq 1.0949192541, \quad H(\Delta_5 | U' \Delta_5 U'^*) \geq 1.1963669266,$$

while the best bases found give 0.5659197506, 1.0946676581 and 1.1958404880. The flagged unitaries are stored in the file `connes_stormer.json` written by the program.

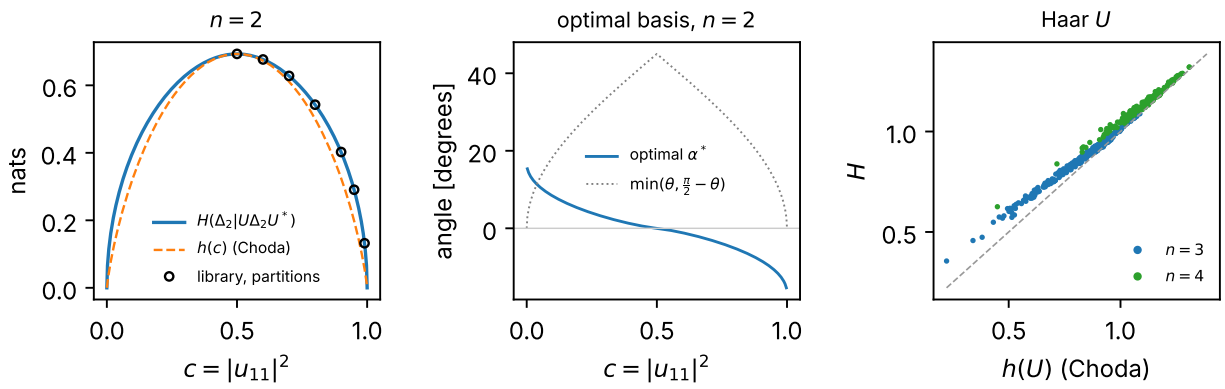


Figure 1: Left: $H(\Delta_2|U\Delta_2U^*)$ (Theorem 4.4) and Choda's value $h(c)$ against $c = |u_{11}|^2$, with the values found by the search over partitions (circles). Middle: the optimal angle α^* of the basis (Proposition 4.6), and $\min(\theta, \frac{\pi}{2} - \theta)$ for comparison. Right: H against $h(U)$ for Haar-random U, $n = 3$ and $n = 4$ (the best of general partitions and bases); dashed: $H = h(U)$.

(iii) *Structure.* In all these cases every element of the best partition has rank one, and the elements point in 4, 5 and 6 distinct directions for $n = 3, 4, 5$, i.e. $n + 1$; by Lemma 2.2(a) the partition is, up to the merging tolerance, a partition with $n+1$ rank-one elements. Two of the directions are far from orthogonal: the largest overlap $|\langle d, d'\rangle|$ is 0.897, 0.980 and 0.991 for the unitaries with the largest gap.

(iv) *Choda's partition is never optimal.* In every sample the best value found exceeds $h(U)$, by at least 0.0002 ($n = 3$; Table 2), as for $n = 2$ (Corollary 5.2). Since this value is attained at an explicit partition, $H(\Delta_n|U\Delta_nU^*) > h(U)$ for every sampled unitary, up to rounding errors. The right panel of Figure 1 shows H against $h(U)$.

(v) *The bound G^* is not attained.* For every sampled unitary the computed G^* exceeds the computed H, by at least 0.004, 0.026 and 0.076 for $n = 3, 4, 5$. Since the computed G^* is attained at an explicit state and the computed H is only a lower bound for H, this is an observation as well: unlike for $n = 2$, the optimal ensembles do not appear to consist of maximisers of G.

Observation 6.1. For $n = 3, 4, 5$ and a small but non-negligible fraction of Haar-random unitaries U, the supremum (1) for $H(\Delta_n|U\Delta_nU^*)$ is not attained at an orthonormal basis: partitions with $n + 1$ rank-one elements are better than every basis found by an extensive multistart search.

Conjecture 6.2. Let $n \geq 3$.

(a) There are unitaries $U \in U(n)$ (in fact, a set of positive Haar measure) with $H(\Delta_n|U\Delta_nU^*) > \max_{W \in U(n)}[h(U^*W) - h(W)]$.

(b) For Haar-almost every U, $h(U) < H(\Delta_n|U\Delta_nU^*) < G^*$.

Remark 6.3 (Non-abelian subalgebras). The same code handles any subalgebra given by an inclusion matrix. For $P = M_2 \oplus \mathbb{C} \subset M_3$ and $P = M_2 \otimes 1 \subset M_4$ against uPu^* for two seeded Haar-random u each, the maxima over partitions with $m = n$, $2n$ and n^2 elements agree to 1×10^{-13} and 3×10^{-6}; we obtain $H = 0.6319, 0.5142$ and $H = 0.6795, 0.6923$, against $G^* = 0.6931, 0.6931$ and $G^* = 0.6931, 0.6931$. For $M_2 \otimes 1 \subset M_4$, $E_P\rho = \mathrm{Tr}_2\, \rho \otimes \frac{1}{2}$ with the partial trace over the second factor,

so $S(E_P\rho) = S(\mathrm{Tr}_2\,\rho) + \log 2 \geq \log 2$, and similarly $S(E_Q\rho) \leq 2\log 2$; hence $G^* \leq \log 2$, a bound that the computed G^* reaches while the computed H stays below it. These values are lower bounds found by the search; we have no closed form.

7 Open problems

(1) *A certificate for Observation 6.1.* Prove Conjecture 6.2(a) for one explicit unitary, e.g. the stored 3×3 unitary with the largest gap, by a rigorous upper bound for $\max_W[h(U^*W) - h(W)]$ over $U(3)$ (a global optimisation over an eight-dimensional manifold modulo phases, amenable to interval branch and bound).

(2) *Closed forms for $n \geq 3$.* Is there a formula for $H(\Delta_n|U\Delta_nU^*)$ for special families, e.g. $U = R_\theta \oplus 1_{n-2}$, unitaries close to permutation matrices, or U close to a complementary one? Is H continuous and piecewise analytic in U, with the switch between bases and partitions with $n+1$ elements as a transition?

(3) *Small angles for $n \geq 3$.* For $n = 2$, H is of first order in the angle θ, while $h(U)$ is $O(\theta^2 \log \frac{1}{\theta})$ (Theorem 5.1). Is $H(\Delta_n|U\Delta_nU^*)$ of first order in the distance of U from the monomial unitaries for every n, and with which constant?

(4) *The number of elements.* Is the supremum for two masas always attained by a partition with at most $n+1$ elements of rank one? Where partitions beat bases, the best partitions found have rank-one elements in $n+1$ directions, and otherwise a basis (n elements) is as good as the best partition found; Proposition 2.3(b) only guarantees n^2 elements.

(5) *The gap to G^*.* Characterise the pairs P, Q with $H(P|Q) = G^*$ (Proposition 3.1); for masas, is equality restricted to $n = 2$ and to complementary or commuting pairs?

(6) *Spin model subfactors.* Is the bound of Remark 5.3 for $H(R_u|R_v)$ sharp, i.e. does $H(R_u|R_v) = H(\Delta_2|u^*v\Delta_2v^*u)$ hold?

(7) *Non-abelian subalgebras.* Closed forms for the examples of Remark 6.3, starting with $M_2 \otimes 1 \subset M_4$.

Code availability. The library, the program that computes all numbers, tables and the figure, and the tests (`tests/test_connes_stormer.py`) are available at `https://github.com/andre-a-no/von_neumann_type_i` (MIT licence).

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